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# Chapter 18
## Lessons and Values: The [0,1] Game Redux

At the outset of this book, we described an approach to studying poker that focused on the dimensions of the game; instead of attempting to solve all of poker all at once, an impossible task, we instead focus on the different elements and characteristics of games and attempt to understand how they affect our play, whether optimal or exploitive. The [0,1] game is a clear example of this; over the course of Part III, we have investigated twelve different [0,1] games, each with different betting structures and rules. To this point, we have focused on the solutions to games, and also to the methodologies we could use to solve any game of this type.

However, another important aspect of analyzing games and learning lessons is to consider the value of the game. The ***value*** of the game is the expectation of a particular player assuming the game is played with ***optimal strategies***. By convention, we state the value of the game in terms of Y's expectation; we do this because almost all poker games favor the second player because of his positional advantage. Thus, when we use this convention, values of games are usually positive. We also, as usual, discuss the ***ex-showdown*** value of the game unless otherwise stated. We often use the letter *G* to denote the game value, whether in a particular region or overall.

We can always calculate the value of a game by brute force; simply calculating all the possibilities for hands that either player would hold, finding the strategies they would employ, the joint probabilities of all possible holdings, and adding up the resultant value distribution. This is often an easy and useful method for solving simple games, such as [0,1] Game #1 (Example 11.2).

| X's hand | Y's hand | Probability | Outcome | Weighted Factor |
|---|---|---|---|---|
| [0, $^1/_2$] | [0, $^1/_2$] | $^1/_4$ | 0 | 0 |
| [0, $^1/_2$] | [$^1/_2$ ,1] | $^1/_4$ | 0 | 0 |
| [$^1/_2$, 1] | [0, '$^1/_2$] | $^1/_4$ | +1 | + $^1/_4$ |
| [$^1/_2$, 1] | [$^1/_2$, 1] | $^1/_4$ | 0 | 0 |
| Total | | | | + $^1/_4$ |

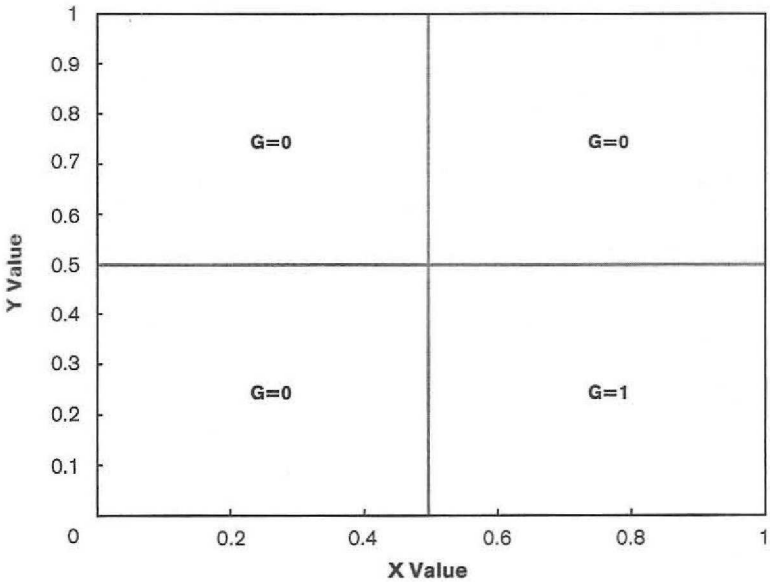


**Figure 18.1 -Game Value for [0,1 ] Game #1**



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So Y's equity in the half-street no-fold game is $^1/_4$. We can apply this methodology to many of the other games we solved as well. However, this method becomes more and more difficult to implement, as the game gets more and more complex. Fortunately, we can take advantage of some of the properties of optimal strategies to find methods that are more suitable for finding game values for games that are more complex.

Consider this same game. Suppose that we hold Y's strategy constant - that is, Y plays his optimal strategy. We can move X's strategy around, knowing that X can never gain value by doing this; remember, if X can improve his equity, then Y isn't playing optimally.

So we can have X follow a "best response" strategy instead of his optimal strategy. This is convenient for us, because on hands where X is indifferent to different actions, we can choose whichever action is easiest for calculation purposes. *Note that this does not mean that X can play these strategies and achieve these values; Y could exploit him.* We are trying to calculate the value of the game here, not the strategies both players will follow.

Considering full-street games:

[0,1] Game #4 (Example 15.1) was more of an instructive examination of degeneracy than a game in the mold of the other games; its equity is quickly calculated by noting that both players put in one bet with all hands. Hence the ex-showdown value of the game is zero.

[0,1] Game #5 (Example 15.2) was the no-fold game with a single raise allowed, but no check-raising. The solution to this game was:

$x_1 = {}^1/_4$
$y_1 = {}^3/_4$
$y_2 = {}^1/_4$

We can easily calculate the equity of this game by holding Y's strategy constant, and finding a best-response strategy for X. The key to the best response strategy is that X can either check or bet between $^1/_4$ and $^3/_4$ against the optimal strategy and obtain the same value. If he bets, he gains a bet from Y's hands between $^3/_4$ and 1, but loses a bet to Y's hands between 0 and $^1/_4$ (compared to checking). Since the equities from betting or diecking are the same, we can simply use + $^1/_4$ as Y's equity when X is on [$^1/_4$, $^3/_4$].

| **X's hand** | **Y's hand** | **Probability** | **Outcome** | **Weighted Factor** |
|---|---|---|---|---|
| [0, $^1/_4$ ] | [0, $^1/_4$] | $^1/_{16}$ | 0 | 0 |
| [0, $^1/_4$] | [$^1/_4$, 1] | $^3/_{16}$ | -1 | - $^3/_{16}$ |
| [$^1/_4$, $^3/_4$] | [0, 1] | $^1/_2$ | + $^1/_4$ | + $^1/_8$ |
| [$^3/_4$, 1] | [0, $^3/_4$] | $^3/_{16}$ | +1 | +$^3/_{16}$ |
| [$^3/_4$, 1] | [$^3/_4$, 1] | $^1/_{16}$ | 0 | 0 |
| **Total** | | | | + $^1/_8$ |



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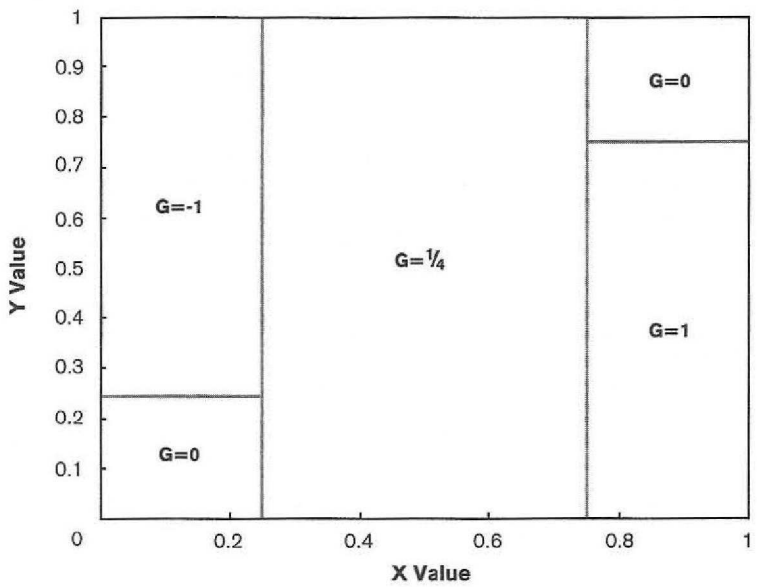


**Figure 18.2. Game Value for [0,1] Game #5**

So the value of the two-bet no-check-raise game to Y is $^1/_8$ of a unit. This is actually quite a bit of value! To put this into terms that you might be more familiar with, let's consider the total action that Y puts into the pot.

When X bets, Y raises Vi of the time, and calls a bet the rest of the time, for a total of $(^5/_4)(^1/_2)$ action, or $^5/_8$ of a bet. When X checks, Y puts in one bet $^3/_4$ of the time, for a total of $(^3/_4)(^1/_2) = {}^3/_8$ of a bet. So the total average action Y puts in from one trial of this game is one bet. For this, he obtains an eighth of a unit, for an edge of 12.5% on his total action.

The reason for this edge is entirely positional; the game is symmetric except that Y acts second. You can see from this simple example that the value of position is inherent in poker. However, this game is not as biased against X as the half-street game was. Recall that in that game, X is reduced to checking and calling. Here, he gains an additional strategic option, that of betting. Y, at the same time, gains the option of raising. It turns out that this game, where X can bet, is worth half as much to Y as the game where X is compelled to check in the dark.

We can further remove the bias against X (in a manner more like regular poker) by looking at the same game, but with check-raise allowed. [0,1] Game #7 (Example 16.4) is the same as [0,1] Game #5 (Example 16.2), except for the addition of check-raise. We should then expect that X will do better because he has gained an additional strategic option.

Let's calculate the equity of this game:

The solution to this game was:

$x_1 = {}^5/_9$
$y_1 = {}^2/_3$
$x_2 = {}^1/_9$
$y_2 = {}^1/_3$

Again, we hold Y's thresholds constant, and examine X's equity from his best response strategies.

When X is below $^1/_3$, we know that no matter what X does, two bets will go in when Y is between 0 and $^1/_3$. X gains a bet from check-raising (compared to betting) when Y is between $^1/_3$



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and $^2/_3$, but loses that bet when Y is between $^2/_3$ and 1. So X is indifferent to betting or check-raising on the entire interval [0, $^1/_3$].

When X is between $^1/_3$ and $^5/_9$, he no longer can profitably check-raise, because the additional bet he loses to Y's good hands is not made up by the additional bet he gains from Y's mediocre betting hands. Hence, he will simply bet in this interval. Above $^5/_9$ X checks.

So we can evaluate the table assuming that X plays the following strategy:

Bet hands below $^5/_9$

Check and call with hands above $^5/_9$.

This strategy has the same equity as X's optimal strategy against Y's optimal strategy. Y could exploit it, but remember we are only trying to find the game value here, not strategies for the two players to follow.

| X's hand | Y's hand | Probability | Outcome | Weighted Factor |
|---|---|---|---|---|
| [0, $^1/_3$] | [0, $^1/_3$] | $^1/_9$ | 0 | 0 |
| [0, $^1/_3$] | [$^1/_3$, 1] | $^2/_9$ | -1 | -$^2/_9$ |
| [$^1/_3$, $^5/_9$] | [0, $^1/_3$] | $^2/_{27}$ | +2 | +$^4/_{27}$ |
| [$^1/_3$, $^5/_9$] | [$^1/_3$, $^5/_9$] | $^4/_{81}$ | 0 | 0 |
| [$^1/_3$, $^5/_9$] | [$^5/_9$, 1] | $^8/_{81}$ | -1 | -$^8/_{81}$ |
| [$^5/_9$, 1] | [0, $^5/_9$] | $^{20}/_{81}$ | +1 | +$^{20}/_{81}$ |
| [$^5/_9$, 1] | [$^5/_9$, $^2/_3$] | $^4/_{81}$ | +$^3/_4$ | +$^1/_{27}$ |
| [$^5/_9$ ,1] | [$^2/_3$, 1] | $^4/_{27}$ | 0 | 0 |
| Total | | | | +$^1/_9$ |

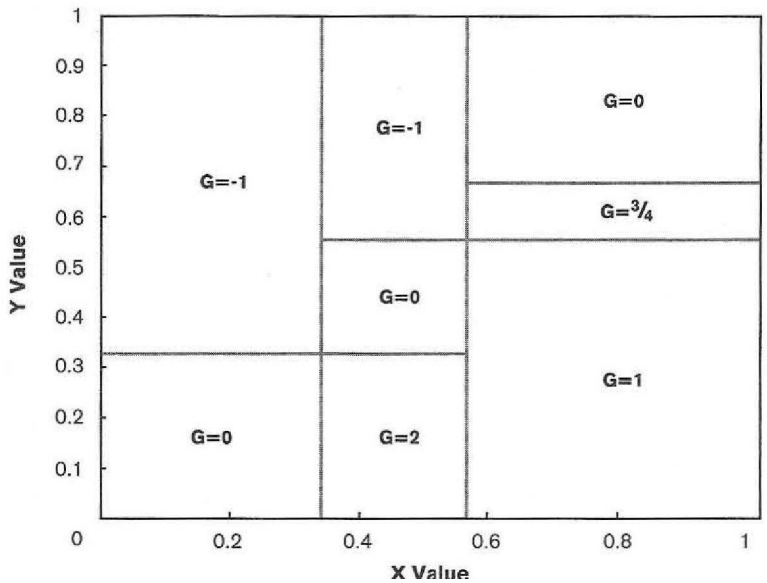


**Figure 18**.3 **Game Value for [0,1] Game #7**

The equity of the game with check-raise is $^1/_9$ compared to the equity of the game without check-raise, which is $^1/_8$. So X gains $^1/_{72}$ of a unit in value from the strategic option of check-raising. This isn't a lot. In fact, when we first solved this game, we expected that the value of check-



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raising would be significant; we conjectured that it would reduce the game value by half. But looking at this more critically, we find that check-raise doesn't really impact Y's strategy tremendously. He raises a little more loosely (from $^1/_4$ to $^1/_3$), and bets a little tighter (from $^3/_4$to $^2/_3$), but neither of these changes create a very large movement in equity from the non-check-raise case.

The final two no-fold games that we considered were the games with infinite raises allowed. [0,1] Game #6 (Example 16.3), where check-raise was not allowed, is the simpler of these games. To find the value of this game, we can take advantage of the game's symmetry once we reach the second bet.

The solution to this game was:

$x_1 = 1/(1 + 2\ r)$
$x_n = r_{n-1}x_1$ (for odd x > 1)
$y_1 = (1 + r)x_1$
$y_n = r_n - lx_1$ (for even n > 1)

Let's consider first the various regions of interest, working backwards from the weakest hands. We can construct a matrix as follows (X's hands across the top, Y's on the left):

| | $[0, x_1]$ | $[x_1, y_1]$ | $[y_1, 1]$ |
|---|---|---|---|
| $[0, x_1]$ | * (see below) | +1 | +1 |
| $[x_1, y_1]$ | -1 | 0 | +1 |
| $[y_1, 1]$ | -1 | 0 | 0 |

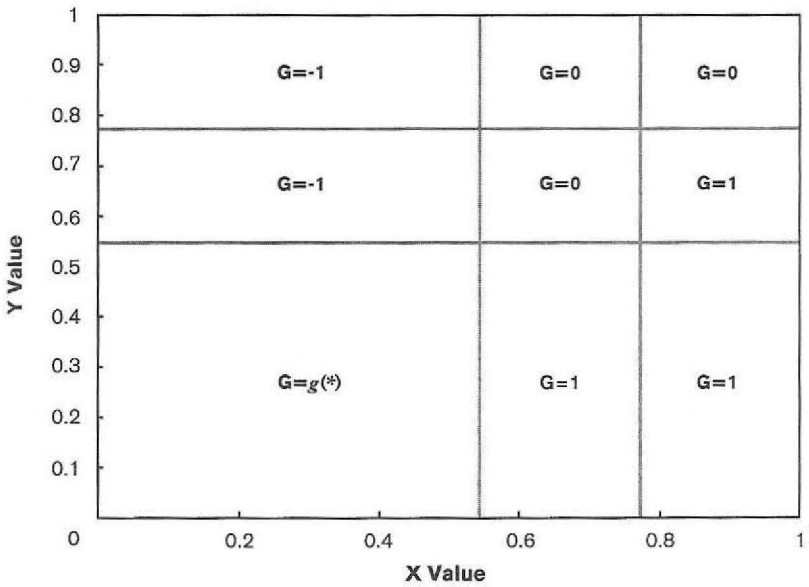


**Figure 18.4. Game Value for [0,1] Game #6**

We have marked the area where both players have a hand better than $x_1$ with a* because the equity in that area is unclear; there are many thresholds where players win different amounts of bets. Specifying the entire range of infinite thresholds would be impossible. However, consider the situation where both players have a hand on $[0, x_1]$. X would bet, and when both players are on this interval, we have a sub-game.



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Let's use g to represent the equity for Y in the region marked *

Within the * region, we have:

| X's hand | Y's hand | Probability | Outcome | Weighted Factor |
|---|---|---|---|---|
| $[y_2, x_1]$ | $[y_2, x_1]$ | $(x_1 - y_2)^2$ | 0 | 0 |
| $[y_2, x_1]$ | $[0, y_2]$ | $(y_2)(x_1 - y_2)$ | +2 | $2(y_2)(x_1 - y_2)$ |
| $[0, y_2]$ | $[y_2, x_1]$ | $(y_2)(x_1 - y_2)$ | -1 | $-(y_2)(x_1 - y_2)$ |
| $[0, y_2]$ | $[0, y_2]$ | $y2^2$ | -g | $-gy2^2$ |

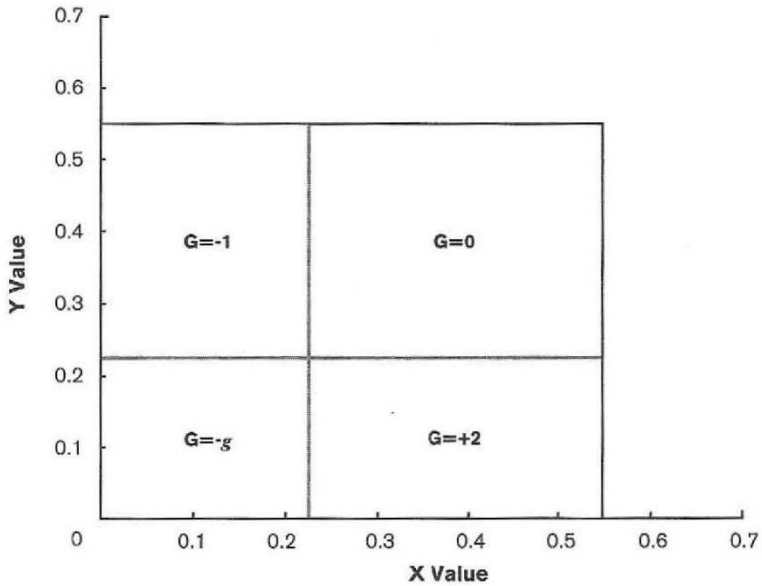


**Figure 18.5. Game Value for [0,1] Game #6 in special region**

After one more raise, this game is the same as in [0,1] Game 6, except that the situation has been reversed! Both players have a hand on a particular interval, and the player who made the last raise has equity -g.

We can now find g:

$g = y_2(x_l - y_2) - gy2^2$
$g = rx_1(x_1 - rx_3) - gr^2x_1^2$
$gr^2x_1^2 + g = rx_1(x1 — rx_1)$
$gr^2 + g = r - r^2$
$g(1+r^2) = r - r^2$
$g(2(1 - r)) = r - r^2$
$g = r/2$

So the equity in this region where both players have strong hands is r/2.

Filling this value into our matrix, we have:

| | $[0, x_1]$ | $[x_1, y_1]$ | $[y_1, 1]$ |
|---|---|---|---|
| $[0, x_1]$ | r/2 | +1 | +1 |
| $[x_1, y_1]$ | -1 | 0 | +1 |
| $[y_1, 1]$ | -1 | 0 | 0 |



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Taking the weighted average equity of the game, we have:

$\langle Y \rangle = (1 - y_1)(y_1 - x_l) + rx_1^2/2$
$\langle Y \rangle = (1/4)(1/4) + r(1/8)$
$\langle Y \rangle = 1/16 + r/8$
$\langle Y \rangle \approx 0.11428$

We can compare this value to the value $1/8$ obtained from the first game in this chapter, where only two bets were allowed without check-raise. This game is slightly more favorable for X than that one. This is of course due to the fact that X can reraise after Y raises. Even then, Y can put in the fourth bet, X the fifth, and so on, but each succeeding bet has a much smaller impact on the EV. The relatively small change that the addition of infinite bets makes on X's equity can be seen by calculating $x_3 = r^2/(l + 2r) = 0.09384$. In order for X to put the third bet in, Y must be above $y_2$, which occurs only about 22% of the time and X must be above $x_3$ a parlay of just 2%.

The last of the no-fold games is the infinite bet game with check-raise allowed (see Example 16.4). We can solve this game in a like manner. Recall that in the two-bet check-raise game, X was indifferent between check-raising and betting with all the hands stronger than $y_2$ in the previous check-raise game. It turns out that this remains the case in the infinite bet game; X is indifferent between check-raising and betting between any two Y thresholds. Hence as our best response strategy for X we will play the simple strategy:

$x_l = y_1$
$x_3 = y_3$

... and so on.

In the region where both players are better than $y_1$, we again find (as in the last game) that the equity of that region is r/2. We invite you to verify that this is the case, as well as verifying X's indifference between his various strategy options.

This creates a simple matrix like the previous one:

| | $[0, y_1]$ | $[y_1, 1]$ |
|---|---|---|
| $[0, y_1]$ | r/2 | +1 |
| $[y_1, 1]$ | -1 | 0 |

Taking the weighted average equity of this game, we have:

$\langle Y \rangle = ry_1^2/2$
$\langle Y \rangle = r/4$
$\langle Y \rangle \approx 0.10355$

Again, we see that the addition of check-raise benefits the first player only a small amount, moving Y's equity from about 0.114 to about 0.104. By contrast, allowing X to bet immediately decreased Y's equity from $1/4$ to $1/8$. We will return to comparing the values of these games shortly, after quickly considering the equity of the finite pot games.

The simplest finite pot game we considered was the half-street game where folding is allowed, [0,1] Game #2 (Example 11.3), whose solution was as follows:



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$y_1 = (1 - \alpha)/(2 - \alpha)(\alpha + 1)$
$x_1^* = 2y_1$
$1 - y_0 = \alpha y_1$

To solve this game, we will hold Y's strategy constant and find a best response strategy for X. We will use the notation $G_n$ to represent the value in regions beginning with the weakest (n = 0) and proceeding to the strongest.

When X is on $[0, y_1]$, he obviously achieves maximum value by calling, since he even beats some of Y's value bets. So here X calls. He breaks even when Y has a value betting hand, and gains one bet against Y's bluffs.

$G_2 = \langle X \text{ on } [0, y_1] \rangle$
$G_2 = -\alpha y_1$

When X is on $[y_1, y_0]$, he is actually indifferent with all these hands to calling or folding. He wins against bluffs and loses against value bets. Since he is indifferent, we can arbitrarily assume that he folds. Then Y gains nothing with value bets and wins the pot of P units with bluffs.

$G_1 = \langle X \text{ on } [y_1, y_0] \rangle$
$G_1 = P\alpha y_1$
$G_1 = (1- \alpha)y_1$

When X is on $[y_0, 1]$, he folds, and Y wins the pot. Y gains half a pot when he is on the same interval as X, so:

$G_0 = \langle X \text{ on } [y_0, 1] \rangle$
$G_0 = P\alpha y_1/2$
$G_0 = y_1(1- \alpha)/2$

To find the overall equity of the game, we can take the weighted sum of these over the interval lengths:

$G = y_1G_2 + (y_0 - y_1)G_1 + (1 - y_0)G_0$
$G = y_1(-\alpha y_1) + (1 - y_1 - \alpha y_1)(1 - \alpha)y_1 + \alpha y_1^2(1 - \alpha)/2$

$$G = (1 - \alpha)y_1 - (2 - \alpha)(1 + \alpha)y_1^2/2 \qquad \textbf{(18.1)}$$

We can further simplify this by using the value of $y_1$ and obtain:

$G = y_1(1 - \alpha)/2$

[0,1] Game #9 (Example 17.1) was the finite pot game for a full street. The game value can be found through a similar process. In the region $[0, y_1]$, we can find that the value of X's checks or bets against Y's optimal strategy is the same:

| **Y's hand** | **X checks** | **X bets** |
|---|---|---|
| $[0, x_1]$ | +1 | +1 |
| $[x_1, y_1]$ | -1 | -1 |
| $[y_1, y_1^*]$ | 0 | -1 |
| $[y_1^*, y_0]$ | 0 | 0 |
| $[y_0, 1]$ | - 1 | 0 |



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*The solution to this game (for Y) was:*

$y_1 = (1 - a)/(1 + a)$
$y_0 = (1 + a_2)/(1 + a)$
$y_1^* = 1 – a$

We see that Y gains a bet in the region $[y_1, y_1^*]$ when X checks, but loses a bet in the region $[y_0, 1]$, We can show that these regions are equally sized as follows:

$y_1^* - y_1 = (1 - a) – (1 - a)/(1 + a) = (a – a_2)/(1 + a) = ay_1 = 1 – y_0$

So X is indifferent to checking or betting against Y's optimal strategy with a hand on $[0, y_1]$. This is convenient as it allows us to reuse all the work from the previous game - the value of this game is equivalent to the value of Game #2 (Example 11.3) where X checks dark except that the value of $y_1$ is different.

Equation 18.1 holds here:

$G = (1 - a)y_1 - (2 - a)(l + a)y_1^2/2$ **(18.1)**

Instead of substituting the value of $y_1$ from the original game, we substitute the new value of $y_1$ from this game:

$y_1 = (1 - a)/(l + a)$

$G = (1 - a)^2/(l + a) - (2 - a)\ (1 - a)^2/2(l + a)$ **(18.1)**
$G = (1 - a)^2(2 - 2 + a)/2(l + a)$
$G = a(l - a)^2/2(l + a)$

The third finite-pot game we looked at was the game where Y could raise but there was no check-raise (Example 17.2). For this game, we can see that the equations for the following values are the same as in the one-bet case:

$G_1 = <X \text{ on } [y_1,y_0] >$
$G_1= Pay_1$
$G_1 = (1 – a)y_1$
$G_0 = <X \text{ on } [y_0, 1] >$
$G_0 = Pay_1/2$
$G_0 = y_1( l - a)/2$

For $G_2$, the case where X is on $[y_1, y_2]$ we can examine X's options. We know that if X bets in this region, he is indifferent to his choices when Y raises. Y will always have him beat with value raises, and X will always beat bluffs. So he is indifferent between bet-folding and bet-calling in this region. Comparing the value of X check-calling in this region to his bet-calling, we have:

| **Y's hand** | **X check-calls** | **X bet-calls** |
|---|---|---|
| $[0, y_2]$ | +1 | +2 |
| $[y_2, y_1]$ | 0 | 0 |



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| | | |
|---|---|---|
| $[y_1, y_1^*]$ | 0 | -1 |
| $[y_1^*, y_2\#]$ | 0 | -2 |
| $[y_2\#, y_0]$ | 0 | 0 |
| $[y_0, 1]$ | -1 | 0 |

Again, we can show that X is indifferent between these choices as follows. One of the indifference equations from this game was:

$y_1^* - y_1 = (P + 1)(y_2\# - y_1^*) + (1 - y_0)$

For bet-calling, X gains:

$2(y_2\# - y_1^*) + y_1^* - y_1 = (P + 3)(y_2\# - y_1^*) + 1 - y_0 = y2 + ay_1$

From check-calling, X gains:

$(y_2 - 0) + (l - y_0) = y_2 + ay_1$

Hence on the interval $[y_2, y_1]$ X is indifferent between check-calling, bet-folding, and bet-calling. To those of you who are interested in solving additional [0,1] games, this type of relationship (which we call a ***triple in***difference) is common in more complex finite pot games, and is often helpful in solving these games with fewer and simpler indifference equations.

We can most easily find Y's equity when X is on this interval by assuming X checks and calls. Then we have:

$G_2 = \langle X \text{ on } [y_2, y_1] \rangle = y_2 - ay_1$

With hands stronger than $y_2$, X's best option is to bet-call. By doing so, he breaks even when Y raises for value, gains one bet when Y calls, and gains two bets when Y raise-bluffs. Y's equity, then, is:

$G_3 = \langle X \text{ on } [0, y_2] \rangle - (y_1^* - y_2) - 2(y_2\# - y_1^*)$
$G_3 = y_2 - y_2\# - (y_2\# - y_1)$
$G_3 = y_2 - (l - a) - a_2y_2$
$G_3 = (1 - a_2)y_2 - 1 + a$

Summing the weighted values of these four regions, we have:

$\langle Y \rangle = y_2G_3 + (y_1 - y_2)G_2 + (y_0 - y_1)G_1 + (1 - y_0)G_0$
$\langle Y \rangle = y_2G_3 + (y_1 - y_2)G_2 + (y_0 - y1)G_1 + (1 - y_0)G_0$
$\langle Y \rangle = y_2(G_3 - G_2) + y_1G_2 + (y_0 - y1)G_1 + (1 - y_0)G_0$
$\langle Y \rangle = -(1 - a)y_2 - a2y_2^2 + (1 + a)y_1y_2 + (1 - a)y_1 - (1 + a)(2 - a)y_1^2/2$

For [0,1] Game #11 (Example 17.3), the check-raise game with two bets, we have an analogous situation to the no-fold game. Holding Y's strategy constant, we can show that X is indifferent to check-raising or bet-calling in the region $[0, y_2]$. Hence the value of Game #11 follows the same formula as the value of Game #10; of course the values of $y_1$ and $y_2$ are different, and hence Y's equity is decreased.

We're going to stop deriving the game values here; as you can see, the values and the process of deriving them scale in complexity along with the games. However, these methods can be used to



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find the value of any game, so we encourage you to attempt to find the values and solutions to other [0,1] games.

## *Lessons of the [0,1] Game*

Now, let's take a deep breath, and review what all these values mean.

Take a look at the following figure, which shows some of the relationships between these games:

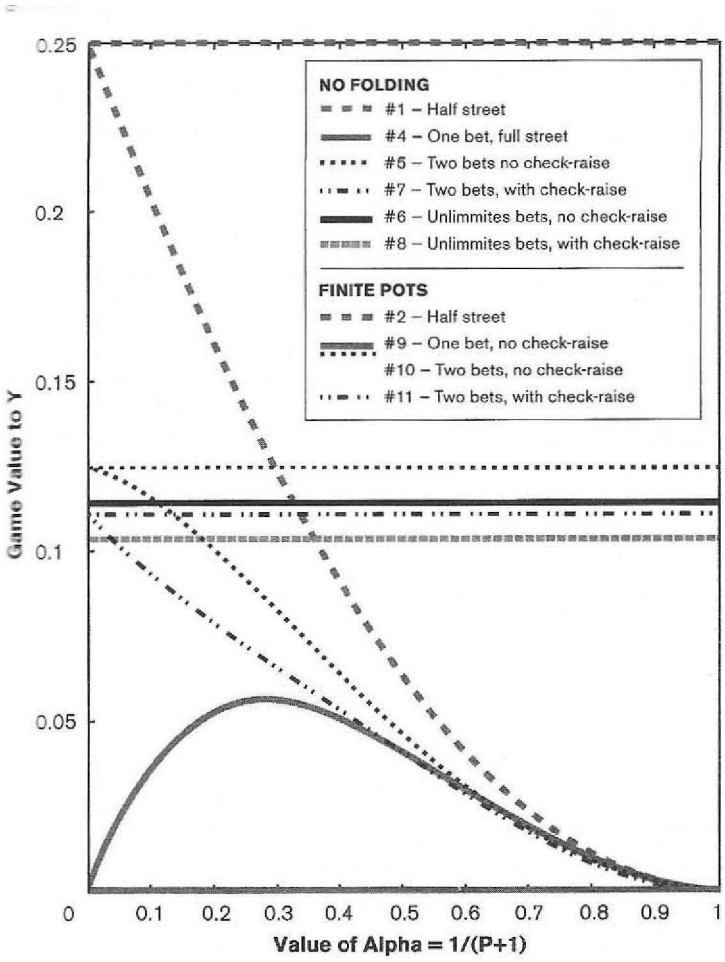


**Figure 18.6. Game Value for various [0,1] Games**

We can see in this figure that some of these games are in fact special cases of each other; for example, Game #1 is actually a special case of the more general Game #2. To obtain Game #1, we simply let the pot become arbitrarily large. This means that the value of a goes to zero. This relationship holds between any finite pot and its no-fold counterpart; we can simply find the value of the game for the finite pot game and take the limit as the pot goes to infinity to find the value of the no-fold game. There are other relationships between the games as well. We can see that some games are obtained by adding strategic alternatives to a different game. For example, Game #7 is simply Game #5, but we give X the opportunity to check-raise. By comparing the equities of the games, we can see the verification of a principle we stated earlier:
*Strategic options have non-negative value.*
Throughout these games, we find examples of this. Consider the half-street games #1 and #2. In Game #2, the primary change that occurs is that X gains the strategic option of folding to Y's bet. Hence the value of Game #2 (from Y's perspective, remember) is reduced. Likewise for all of the following transitions:

Between Game #1 and Game #4, where X gained the option to bet.



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Between Game #4 and Game #5, where Y gained the option of raising.
Between Game #5 and Game #6, where X gained the option of reraising.
Between Game #5 and Game #7, where X gained the option of check-raising.

In all of these cases, and in their counterparts in the finite pot games, the player who gained a strategic option gained (or at least retained their previous) equity.

The second principle that we find in the [0,1] game is the value of *r*. In real poker, there are card removal effects, the effects of multi-street play, and so on that subtly change the value of *r*. But the idea that you should raise your opponent on the river with about 41% of the hands that he would bet is a practical one.

Consider the following game. Suppose you and a friend decide to play a variant of holdem where you each receive two cards, but then the entire board is dealt, and there is one round of betting (the river). Suppose you structure the betting so that Y can either bet or show down, then X can raise and you allow unlimited raises. The optimal strategy for this game is something roughly like this: the first players bets r of his hands; the second player raises with $r^2$ of the hands, the first player reraises with $r^3$ of the hands, and so on. Suppose the board was J♥ 9♣ 8♠ 3♦ 2♣. Then the hands could be ranked from top to bottom by value. Each player can have a total of 1,081 hands.

| **Hand** | **Number of Ways** | **Cumulative # of ways** |
|---|---|---|
| QT | 16 | 16 |
| T7 | 16 | 32 |
| JJ, 99, 88, 33, 22 | 30 | 62 |
| J9, J8 | 18 | 80 |
| J3 or worse two pair | 72 | 152 |
| AJ, KJ, QJ | 36 | 188 |
| JT, J7, J4 | 60 | 248 |
| A9-Q9, T9, 97-94 | 96 | 344 |
| A9-Q9, T9, 97-94 | 96 | 440 |
| Other hands | 641 | 1081 |

$(1081)r \approx 448$
$(1081)r^2 \approx 185$
$(1081)r^3 \approx 77$
$(1081)r^4 \approx 32$
$(1081)r^5 \approx 13$

In doing true optimal strategics for this game, we would be forced to consider card removal effects and account for the fact that the hands are not uniformly distributed, etc. Obviously, you would never stop raising with the nuts in real poker, and so on. But equally, we can see that in the above example, the approximate cutoffs for putting in the $n^{th}$ bet are third pair or better for the first bet, QJ or better for the second bet, first and third pair for the third bet, the second nut straight for the fourth bet, and the nuts for the fifth.

Can we apply this directly to holdem? Not precisely, because the play on previous streets and the resultant distribution of hands changes how the hands are allocated. But using this guideline of 41% (if your opponent cannot, fold - for example the pot is too big and he cannot have a draw) is



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a reasonable guess. In games where the opponent can fold, we can adjust this number downward (because the opponent still raises with his best hands but can now fold his worst), to about 35%.

This last example, in addition to showing how we might apply the value of r to real poker, provides an introduction to the last and most important lesson of the [0,1] game. In all the [0,1] games, hands were partitioned into regions, where each player took various actions. We saw that at the threshold points, the players were indifferent to taking the actions on either side. But they are not indifferent to taking all actions most of the time.

When we calculated some of the game values, we held Y's strategies constant and found that in most of his value regions, X was indifferent to taking various value actions. This means that as a practical matter, if Y plays optimally, X has some latitude to make mistakes near his threshold values without it costing much. We call this type of strategy ***flexible.*** For Y, however, this is not the case. If X plays optimally and Y fails to bet hands a litde better than $y_1$, Y loses equity immediately. We call strategies like Ys ***brittle.*** X's strategies are often flexible in [0,1] games because with hands near his betting threshold, if he should check incorrectly, Y will often put in the bet anyway. Such a situation doesn't exist for Y, however.

When you play a side whose strategies are brittle (which happens frequently to both sides of real poker), the selection of hands with which you take each action is very important for optimal play. Organizing hands into regions is both a logical and practical step as well as a high-expectation practice. What this means is that typically in real poker, we will identify cutoffs between various actions and have clear ideas about what our intent in making such a bet or raise is. On the river, these ideas are primarily centered around how many bets we are willing to put in, and whether we will call or fold to a raise. On previous streets, these decisions are primarily oriented toward whether we are betting for value or semi-bluffing, and how we will respond to future action.

Additionally, this means that we typically do not play mixed strategies with very many hands: that is, at the boundaries there are mixed strategies, and there may occasionally be reason to play mixed strategies for the purpose of hiding card removal effects, but normally hands adjacent in value are played in the same way.

Also to be noted is the fact that the [0,1] strategies for finite pot games are filled with regions that exist primarily to make the strategies balanced. For each value bet, there is a corresponding bluff, and a set of hands within the value region that will call a raise and some that will fold a raise. Thinking about real poker play in this way is a very strong tool for enabling you to play in a balanced manner. For every value bet you make, consider the bluffing region that corresponds. How big is it? Are the ratios proper? When raising, do you raise-bluff? With which hands do you do so? Questions like these are the kind that lead to balanced play that is difficult to exploit.

We close this chapter with a look at the last [0,1] game in this book, which provides an interesting result related to high-low games.

## Example 18.1 - The [0,1] *x* [0,1] Game

In this variant of the [0,1] game, each player receives two real numbers independently and uniformly distributed from 0 to 1. One of these numbers is pre-designated to stand in as his "high" hand, and the other as his "low" hand. Both hands are subject to the usual rules of the [0,1] game; the lower "high'" number gets half the pot and the lower "low" number gets half the



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pot. As a result, each possible pair *(x, y)* where *x* and y axe between 0 and 1 represents a possible hand in this game.

Fox this game we stipulate that neither player may fold, there are two bets left, and check-raise is not allowed. The pot size, as is usual in games where neither side may fold, is irrelevant.

A high percentage of bets in this game axe in fact a type of bet that we haven't discussed yet. These bets are called ***semi-bluffs,*** where a player makes a bet expecting to often win half the pot when called. These bets are extremely common in high-low games, but occur in holdem as well. (As an example consider the play on the river of a two-paired board. Ace-high will often bet expecting to often win half the pot when called by another ace-high.) In high-low games, however, bets of this nature, where a player has a reasonably strong hand one way and a mediocre hand the other make up a substantial part of all bets.

One way of looking at the game is by graphing the game as xy-coordinate pairs in a plane. Instead of a single threshold, however, for each player's strategy, we have a boundary that encloses a region fox an action. We initially parameterize the game as having three curves - we call them $x_1$, $y_1$; and $y_2$ by analogy with [0,1] Game #5 (Example 16.2). We also employ the terms "below" the curve as referring to the area closer to (0,0) and "above" the area closer to (1,1). These three curves, then, are related as follows:

$(0, 0) < y_1 < x_1 < y_2 < (l, 1)$

We have indifference equations at each boundary as well. However, generating these indifference equations is a little more complex than previously. In this case, we will begin by assuming X's strategy first, deriving Y's strategy from that, and then retaining and trying to improve X's equity by varying his strategy to prove that the strategy we assumed for X was in fact optimal.

Assumption: $x_1$ lies on the line $y = 1 - x$.

This assumption may perhaps make intuitive sense to the reader, by analogy with $x_1$ in the one-dimensional [0,1] games that was $^1/_2$ in the no-fold-allowed case.

Next consider the boundary $y_2$. Y must be indifferent between calling (above $y_2$) and raising (below $y_2$). The next step is to identify the regions that generate expectation for either player. Note that in many regions the pot is split and neither player gains. For any point (x, y) on $y_2$:

When X's hand in is a rectangle with vertices {(0, 0), (0, y), (x, 0), (x, y)}, Y loses one bet by calling, but two by raising. This rectangle is the only place where X scoops Y. Y in turn scoops X in a triangular region bounded by the vertices {(x, 1 - x), (1 - y, y), (x, y)}. See Figure 18.7.



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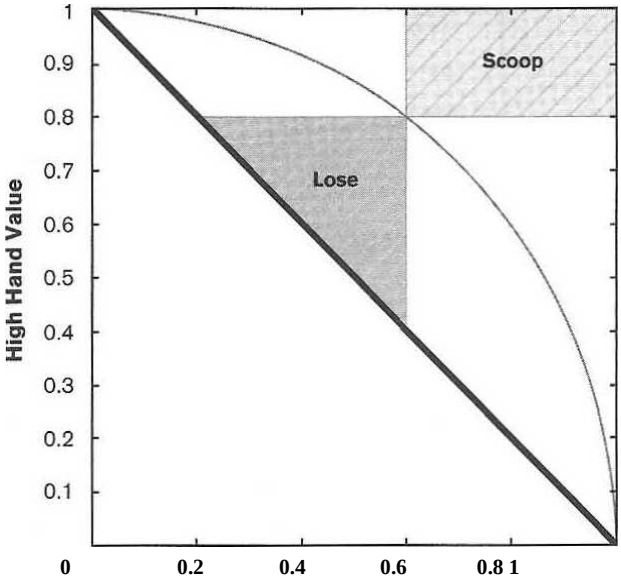


**Figure 18.7. Region comparisons for hands in the [0,1] *x* [0,1]**

Each of these regions is worth one bet, so setting their areas equal:

$xy = {}^1/_2\,(1 - y - x)\,(1 - x - y)$
$2xy = 1 - y - x - y + xy + y^2 - x - x^2 + xy$
$1 = x^2 + y^2 - 2x - 2y + 2$
$l = (x - l)^2 + (y - l)^2$

Students of algebra will recognize this as the equation of a circle of radius 1, centered at (1, 1). So we find that $y_2$ is a quarter-circle of radius 1 centered at (1, 1}.

Turning to $y_1$, we look at Y's indifference between checking and betting; Y gains a bet by betting when X is in a rectangle with vertices {(1, 1), (1, y), (x, 1), (x, y)) and loses a bet when betting in a triangle with vertices {(x, y), (x, 1 - x), (y, 1 - y)}.

Setting the areas of these regions equal:

$(1 - x)(l - y) = {}^1/_2\,(y - (1 - x))(x- (1 -y))$
$1 - x - y + xy = {}^1/_2(y- 1 + x)^2$
$1- x - y + xy = {}^1/_2\,(y^2 - y + xy - y + 1 - x + xy - x+ x^2)$
$2 - 2_x - 2_y + 2_{xy} = y^2 - 2y + 2xy - 2x + x^2 + 1$

$1 = y^2 + x^2$

This is the equation of a circle of radius 1 centered at (0, 0). Plotting these three graphs, we see that the strategies form a "football" about the centerline y = 1 - x.



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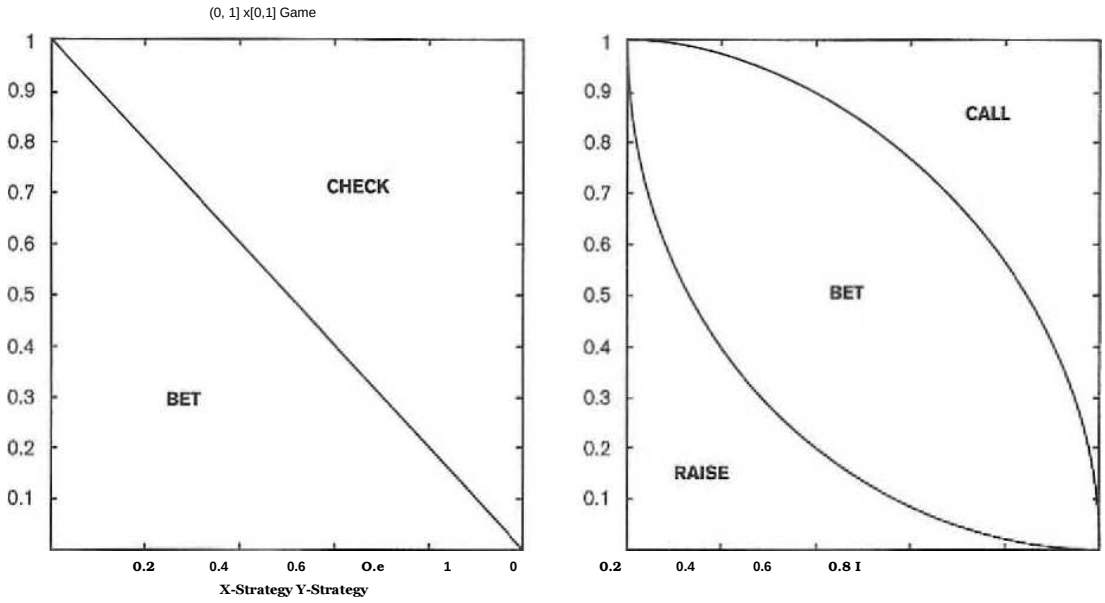


We now return to our assumption and show that $x_1$ does in fact he on the line y = 1 - x. To do this, we can use the fact that we know that along $x_1$, X is indifferent to betting or checking. Along this boundary, X gains a bet by checking (over betting) when Y is below $y_2$, because he does not get raised. At the same time, he loses a bet when Y lies above $y_1$ because he fails to gain a bet from Y. Of course, for some hands above $y_1$ and below $y_2$, there is no difference because the pot is split.

We know that the size of the two regions A and B must be equal. Intuitively we can see that along the diagonal from (0, 1) to (1, 0) there will be symmetry between the two regions. Since the diagonal is actually y = 1 - x, we have our optimal strategies.

This gives rise to at least one interesting answer: if it's a two-bet high-low game with no qualifier, no card removal effects, no check-raise, and neither player may fold, what percentage of hands should the second player bet? The answer is $\pi/4$!

These restrictions are quite onerous and so this game is not a true analogue to real-life high-low games. However, it is at least sometimes the case that when playing high-low games (for example stud), we find ourselves in a position very much like the "no fold" restriction The distribution of hands with which we have arrived at seventh street is such that neither player can fold, due to the high probability that the opponent has a one-way hand. Even weakened in this way, this result indicates a significantly higher betting frequency than is commonly thought, particularly by the player who is acting in last position.

## *Key Concepts*

- Comparing the values of various games gives us insight into the importance of various concepts.
- Adding strategic options to a one player increases the value of the game for that player.
- When considering the value of a game, we can often hold one player's strategy constant and allow the other player's strategy to maximally exploit it in order to simplify the game value calculation.
- Some strategic options are brittle - that is, small deviations from optimal cost equity immediately. Others are flexible – that is, against the optimal strategy, they can move around in a range.
- Strategic options have non-negative value.



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